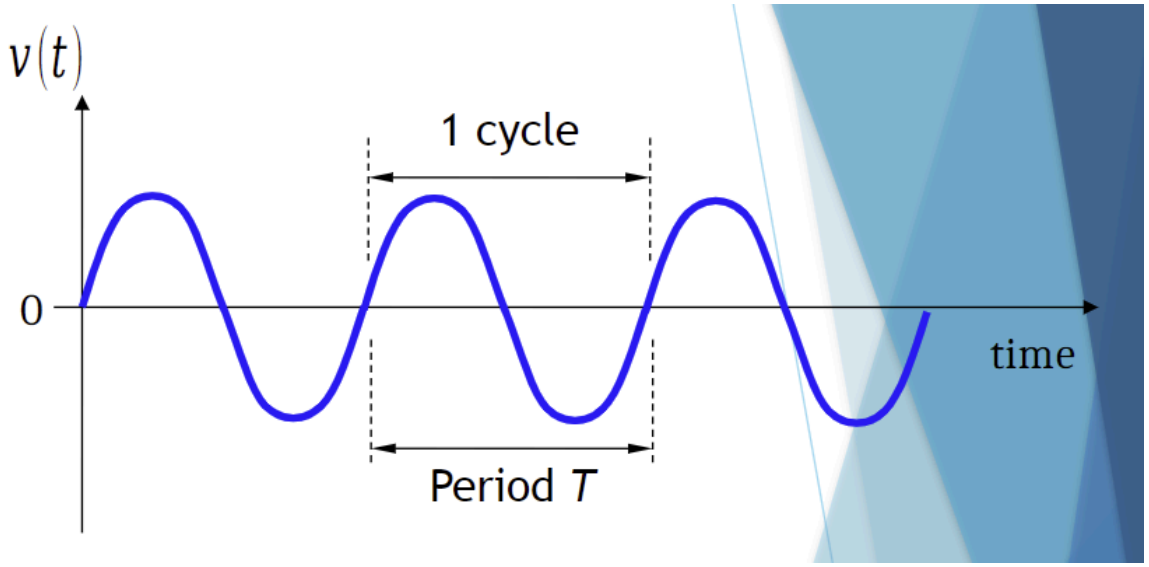
AC Voltage & Current

Alternating refers to changing polarity(for voltage) and direction (for current) at a certain rate respect to time. The most common type of alternating relationship is the sine wave.

Sinusoidal AC voltage

On the graph of voltage against time for a sinusoidal AC, when the wave is above the x axis, the voltage is positive and current flow in a specific direction.

Below the x axis the voltage is negative and the current flows in the opposite direction. This cycle repeats regularly.

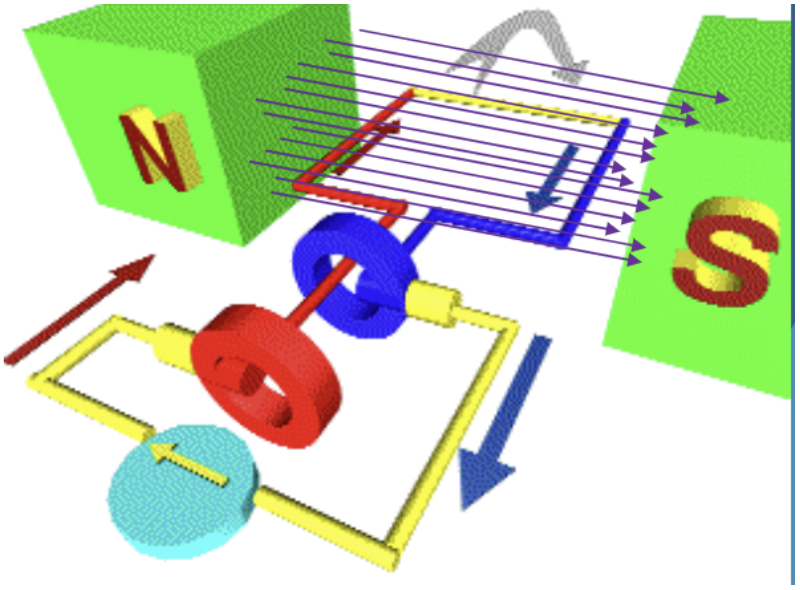


We can talk about how frequent the voltage changes with frequency and period, linked by this equation:

$$f = \frac{1}{T}$$

Note that we use period and not wavelength, as this is dependent on time.

AC generators



An AC generator works by turning a coil of wire in a magnetic field, which causes the wire to cut the magnetic flux lines, causing a changing magnetic field in the coil. This causes a voltage to be induced in the coil, inducing an alternating current through the coil.

Peaks of the AC current is during the coil being parallel to the magnetic field, while when the coil is perpendicular to the magnetic field, the AC current is 0. every 180 degree turn, the direction changes, as the same section of coil changes direction of motion after the turn, thus causing a different direction of current. (can be derived from Flemings Right hand rule)

The frequency of the generated AC is dependent on the rotations per seconds of the coil and the number of pole pairs (aka how many groups of N-S are in the generator.)

the frequency f of the output AC is

$$f = \text{number of pole pairs} \times \text{rotations per second}$$

The amplitude is dependent on the number of coils in the coil and the rotation speed, as by faraday's law, a faster speed causes a higher change in magnetic flux, which increases the induced voltage. Thus,

$$V \propto \text{no. of coils} \times \text{rotations per second}$$

General Wave equation

We can write the general expression of the AC voltage as so:

$$v(t) = V_p \sin(2\pi f t) = V_p \sin(\frac{2\pi t}{T})$$

Vp represents the peak AC voltage, aka the amplitude of the wave.

Sometimes, instead of Vp being given, the value Vpp, or peak to peak voltage is given, which is from +Vp to -Vp. Thus, Vpp = 2Vp.

Sometimes, the wave equations can be written in terms of an angle:

$$v(\theta) = V_p \sin(\theta), \text{where } \theta = 2\pi f t$$

Both degrees and radians can be used for theta. By default, 2*pi*ft is in radians.

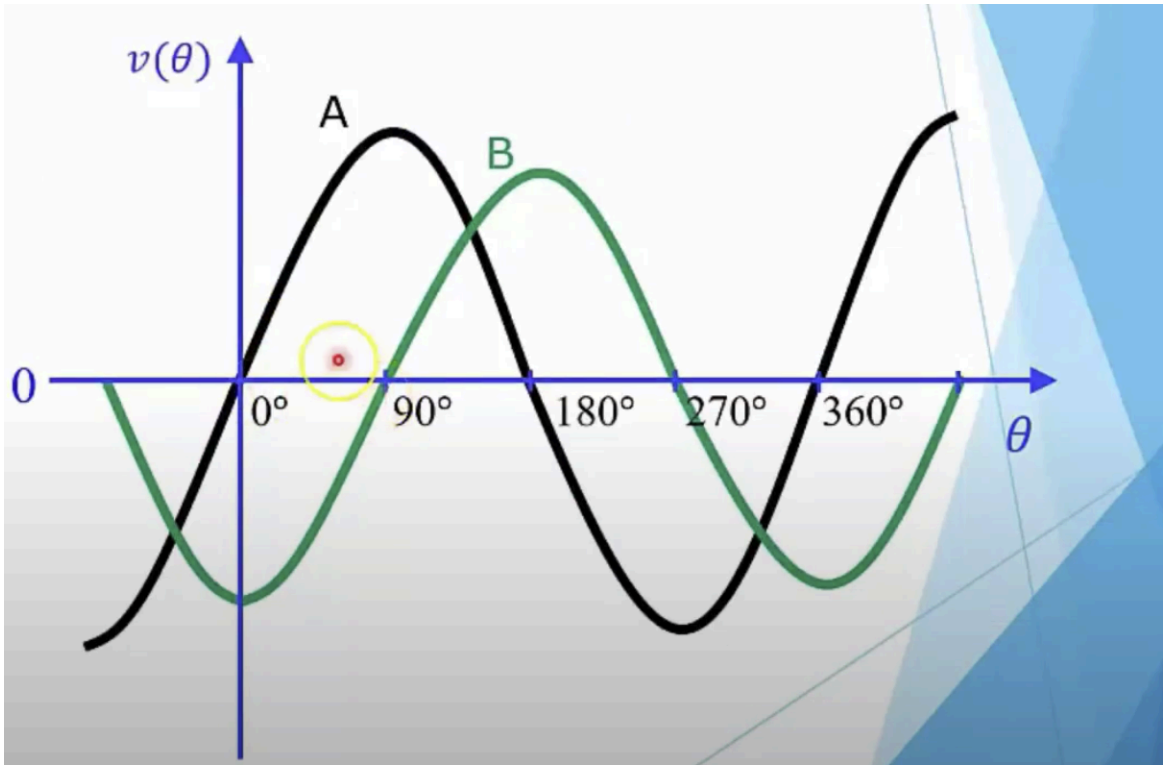
From these functions, we can also find the instantaneous value of the AC voltage/current, by plugging in the current time/angle into the equation to get our answer.

These wave equations may also have a phase angle, ϕ.

For the angle version of our wave equation,

$$v(\theta) = V_p \sin(\theta \pm \phi)$$

Note that +phi shifts the wave backwards while - phi shifts the wave forward.



For two waves with different phases, we can describe their phases by which leads/lags which.

As A rises above the zero level at 0 degrees which is earlier than B which rises at 90 degrees, A leads B (by 90), or B lags A (by 90).

Average and Root Mean Square Values

The average value/voltage of a wave can be thought of as the integral from 0 to 1/2 of the period (using the full period will include the equal positive and negative parts of the wave, thus we restrict to one positive half cycle)

$$V_{avg} = \frac{2}{T}\int_0^{\frac{T}{2}} V_p \sin\left(\frac{2\pi t}{T}\right) dt = \frac{2}{\pi}V_p$$

We can solve the integral as so:

$V_{avg} = \frac{2}{T}\int_0^{\frac{T}{2}} V_p \sin\left(\frac{2\pi t}{T}\right) dt$

$= \frac{2}{T}\left[-V_p(\frac{T}{2\pi})\cos\left(\frac{2\pi t}{T}\right)\right]_0^{\frac{T}{2}}$

$= \frac{2}{T}\left(-V_p(\frac{T}{2\pi})\cos\left(\frac{2\pi(0.5T)}{T}\right) - -V_p(\frac{T}{2\pi})\cos\left(\frac{2\pi 0}{T}\right)\right)$

$= \frac{2}{T}\left(V_p(\frac{T}{2\pi}) - -V_p(\frac{T}{2\pi})\right)$

$= \frac{2}{T}\left(V_p(\frac{T}{2\pi}) + V_p(\frac{T}{2\pi})\right)$

$= \frac{2}{T}\left(2V_p(\frac{T}{2\pi})\right)$
$= \frac{2}{\pi}V_p$

Thus, we can use this formula to find the average value. Note this formula also works for negative half cycles too.

Another value that represents a similar quantity is the root mean square value, calculated as this:

$$V_{rms} = \frac{1}{T}\sqrt{\int_0^T v^2(t)\ dt} = \frac{V_p}{\sqrt{2}}$$

Notice how both V_avg and V_rms are both directly proportional to Vp, not the frequency.

The special thing about the rms value is that it relates to dc voltage closely in terms of power.

If we have a resistor connected to $50V_{rms}$ AC, or $50\sqrt{2}\ V_p$, and the resistors dissipates a certain amount of power P, when connected to a dc voltage at 50V, the power dissipated will be the same.

Thus, we mostly use the rms value for circuit analysis.

**ONE THING TO NOTE, peak, instantaneous and average values may be substituted as current instead of voltage, all formulas still apply.

Circuit Analysis

All the circuit laws from DC (KCL, KVL, VDR, ohms law, CDR) all still apply. The main difference is what types of voltages we are using.

AC only

There are 2 main ways to do circuit analysis, either based on the RMS value or on the peak value.

<u>Values given in AC circuits are always assumed to be RMS value, unless specified otherwise.</u>

Thus, we can use the rms values as normal dc voltages to find voltage, current and resistance. Likewise for peak value, we can compute them similarly.

We can also convert between them if necessary by the formula.

$$I_{rms} = \frac{I_p}{\sqrt{2}}, V_{rms} = \frac{V_p}{\sqrt{2}}$$

AC with DC

If the AC voltage of an AC voltage source is $V = V_p \sin(\theta)$, when the AC voltage is connected in series with a DC voltage of V_0, $V_{max} = V_p + V_0$ and $V_{min} = -V_p + V_0$

Note that now with AC+DC circuits, the circuit may not necessarily be oscillating.

Consider an AC source of $3\sin(\theta)$ connected in series with a DC voltage of 5V.

Thus, the max voltage is 3+5 = 8V and the lowest voltage is -3+5= 2V. Notice how both the min and max voltage is positive, thus the direction does not change. This means that the voltage and current is non alternating.

In general, if Vp < V0, the current and voltage will be non alternating. Otherwise, it will alternate.

(can use peak value to find peak values for loads $\longrightarrow$ hence find rms value for load?)